

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399323
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Wang; Liming et al.

System and method for protecting optical fibre splice

Abstract

The present invention relates to a low-profile splice protection system for protecting multi-fibre fusion splice sites. The splice protection system comprises coating material to package the splice site and may comprise a protective housing.

Inventors: Wang; Liming (Shanghai, CN), Zhang; Xiaodong (Shanghai, CN)

Applicant: COMMScope TELECOMMUNICATIONS (SHANGHAI) CO. LTD. (Shanghai, CN)

Family ID: 1000008781964

Assignee: CommScope Telecommunications (Shanghai) Co. Ltd. (Shanghai, CN)

Appl. No.: 18/599699

Filed: March 08, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240288634 A1	Aug. 29, 2024

Foreign Application Priority Data

CN	201410741182.8	Dec. 04, 2014
----	----------------	---------------

Related U.S. Application Data

continuation parent-doc US 18170083 20230216 ABANDONED child-doc US 18599699
continuation parent-doc US 17060996 20201001 US 11619782 20230404 child-doc US 18170083
continuation parent-doc US 15532885 US 10845540 20201124 WO PCT/CN2015/096315
20151203 child-doc US 17060996

Publication Classification

Int. Cl.: G02B6/255 (20060101)

U.S. Cl.:

CPC G02B6/2558 (20130101);

Field of Classification Search

CPC: G02B (6/255); G02B (6/2551); G02B (6/2558); G02B (6/3825); G02B (6/3885)

USPC: 385/78; 385/95; 385/96; 385/99

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4077702	12/1977	Kunze et al.	N/A	N/A
4254865	12/1980	Pacey et al.	N/A	N/A
4714316	12/1986	Moore	385/99	G02B 6/2835
4778242	12/1987	Ota et al.	N/A	N/A
5102212	12/1991	Patterson	N/A	N/A
5247598	12/1992	Takimoto et al.	N/A	N/A
5367591	12/1993	Seike et al.	N/A	N/A
5416873	12/1994	Huebscher et al.	N/A	N/A
5832162	12/1997	Sarbell	N/A	N/A
6485199	12/2001	Ware et al.	N/A	N/A
6676299	12/2003	Durrant	385/59	G02B 6/3885
9304262	12/2015	Lu	N/A	N/A
10845540	12/2019	Wang et al.	N/A	N/A
11619782	12/2022	Wang et al.	N/A	N/A
2004/0062480	12/2003	Cronk et al.	N/A	N/A
2012/0281951	12/2011	Takahashi	156/60	G02B 6/3846
2013/0251317	12/2012	Faulkner	385/96	G02B 6/255
2014/0321813	12/2013	Lu	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1548492	12/2003	CN	N/A
1668953	12/2004	CN	N/A
1735825	12/2005	CN	N/A
101866034	12/2009	CN	N/A
102177454	12/2010	CN	N/A
202494801	12/2011	CN	N/A
43 30 744	12/1994	DE	N/A
201 06 111	12/2000	DE	N/A
20 2007 013 173	12/2007	DE	N/A
2004-347801	12/2003	JP	N/A
2005-250294	12/2004	JP	N/A

2013/077969	12/2012	WO	N/A
2013/126429	12/2012	WO	N/A
2013/151582	12/2012	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report for corresponding European Patent Application No. 15864740.4 mailed Jul. 23, 2018, 8 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority for corresponding International Patent Application No. PCT/CN2015/096315 mailed Mar. 7, 2016, 12 pgs. cited by applicant

Office Action for European Patent Application No. 15864740.4 mailed May 10, 2022, 7 pages. cited by applicant

Yablon, A., "Optical Fiber Fusion Splicing", Springer Berlin Heidelberg, 5 pgs. (2005). cited by applicant

Primary Examiner: Tavlykaev; Robert

Attorney, Agent or Firm: Merchant & Gould P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. patent application Ser. No. 18/170,083, filed Feb. 16, 2023, now abandoned; which is a Continuation of U.S. patent application Ser. No. 17/060,996, filed Oct. 1, 2020, which is a Continuation of U.S. patent application Ser. No. 15/532,885, filed Oct. 6, 2017, now U.S. Pat. No. 10,845,540, issued Nov. 24, 2020, which is a National Stage entry of PCT/CN2015/096315, filed Dec. 3, 2015, which claims the benefit of Chinese Patent Application No. 201410741182.8, filed Dec. 4, 2014, the disclosures of which are hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

(1) The present invention relates to an optical fibre connector and a method of manufacturing same. In particular, the present invention relates to protection for the splice site within the optical fibre connector.

BACKGROUND ART

(2) Optical fibre communication systems are becoming widespread. In some areas, service providers wish to provide high-bandwidth telecommunication capabilities (such as data and voice) to customers. Optical fibre communication systems use optical cable networks to transmit large volumes of data and voice signals over relatively long distances. Optical fibre connectors are important components of most optical fibre communication systems. The optical fibre connectors allows rapid optical connection or disconnection of two optical fibres.

(3) A typical optical fibre connector includes a ferrule or junction. A spring is used to press the ferrule upward, in a forward direction relative to the connector housing. The ferrule performs the role of supporting the end of at least one optical fibre. (In the case of most multi-fibre ferrules, the ends of multiple optical fibres are supported.) The ferrule has a front-end face, and the polished ends of the optical fibres are positioned at said front-end face. When two optical fibre connectors are connected to each other, the front-end faces of their respective ferrules are adjacent to each other, and the ferrules are pressed together by the spring load of their respective springs. When optical fibre connectors are connected, their respective optical fibres are generally coaxially aligned, with the result that the optical fibre end faces oppose each other directly. In such a manner,

optical signals can be transmitted from optical fibre to optical fibre via aligned end faces.

(4) Types of optical fibre connectors may include direct end-receiving optical fibre connectors and splice-on optical fibre connectors. Direct end-receiving optical fibre connectors have ferrules which directly receive their corresponding optical fibres. Conversely, splice-on optical connectors include ferrules that support optical fibre stubs. The optical fibre stubs are spliced to the corresponding optical fibres of optical cables. Exemplary publications that disclose splice-on connectors include patents with PCT international publication numbers WO 2013/126429 and WO 2013/077969. For splice-on optical fibre connectors, the ability to effectively and efficiently protect splice sites is an important design consideration. In this respect, it would be beneficial to provide a splice protection system which could be easily and quickly installed, that is suitable for small optical fibre structures, and that provides flexible protection for splice sites.

SUMMARY OF THE INVENTION

(5) The guiding principles for the present disclosure are the need for a method and structure for effectively and efficiently protecting optical fibre splice sites within optical fibre connectors. Some aspects of the present invention relate to a splice protection method and structure suited to using splice-on multi-fibre optical connectors. Other aspects of the present invention relates to a method and structure for protecting multi-fibre splices within the connector body of an optical fibre connector providing a stable-environment and a sealed structure. A further aspect of the present invention relates to a low-profile splice protector, which is suitable for protecting a row of optical fibre splices (e.g., the site of multiple fusion splices) and is suitable for lateral mounting within the body of an optical fibre connector. In certain embodiments, the optical fibre splice is a fusion splice delimited between a first group of optical fibres and a second group of optical fibres. In addition, another aspect of the present invention relates to a splice protector, which: a) permits a row of splices between optical fibres to be effectively and laterally mounted within the splice protector; b) has a low profile which makes it easy to laterally mount the splice protector into the connector body; and c) is easily filled with a curable protective material such as adhesive cement, which is for filling the gaps within the splice protector and for stabilising, protecting, and mechanically strengthening the splice site. In one example, the splice protector may have a U-shaped profile when viewed in cross-section.

(6) Another aspect of the present invention relates to a splice protection system for protection of fusion splice sites delimited between multiple first optical fibres and multiple second optical fibres. The splice protection system comprises: a splice protector, said splice protector comprising a sleeve, said sleeve comprising mutually perpendicular length, width, and thickness. The sleeve comprises a first main side and a second main side, which are opposite each other and are delimited by sleeve length and width. The first main side and the second main side comprise separated main side walls separated from each other by an intervening space extending along the thickness of the sleeve. The sleeve further comprises a first longitudinal secondary side and a second longitudinal secondary side positioned opposite each other and delimited by sleeve length and thickness. The sleeve further comprises a first lateral secondary side and a second lateral secondary side positioned opposite each other and delimited by sleeve width and thickness. The first and second lateral secondary sides are open lateral secondary sides. The width of the sleeve is at least twice the thickness of the sleeve. The splice site is disposed inside the sleeve of the splice protector. Said multiple first optical fibres and multiple second optical fibres extend outward from the first lateral secondary side and the second lateral secondary side of the splice protector sleeve. The splice protection system further comprises adhesive cement. The adhesive cement is disposed within the splice protector sleeve in order to fill the gaps within the splice protector sleeve.

(7) In another aspect, the first longitudinal secondary side comprises a longitudinal wall. The longitudinal wall bridges the gap between said main side walls and interconnects said main side walls. Moreover, the first longitudinal secondary side is a closed longitudinal side, and the second longitudinal secondary side is an open longitudinal side. In another aspect, the sleeve could be

formed from two matching half-shells.

(8) Yet another aspect of the present invention relates to a method for protecting a splice site with a splice protector sleeve. Said splice site is delimited between the multiple first optical fibres and the multiple second optical fibres. The splice protector sleeve has a length, a width, and a thickness. The splice protector sleeve comprises a first side wall, a second side wall opposite the first side wall, a third wall connecting the first side wall and the second side wall, and open ends. Said method comprises the steps below: (a) putting the splice site within the splice protector sleeve; (b) injecting adhesive cement into the splice protector sleeve and filling the gaps between the optical fibres and the splice protector sleeve; and (c) curing the adhesive cement.

(9) Various supplementary aspects will be set forth in the description below. Each aspect relates to single features and to combinations of features. Please note that both the foregoing general description and the following detailed description are merely exemplary and illustrative and are not limitations on the broad inventive concepts on which the embodiments disclosed herein are based.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 presents an example of an optical cable connector device suitable for implementing the various aspects of the present invention.

(2) FIG. 2 is an exploded view of the optical cable connector device in FIG. 1.

(3) FIG. 3 is a three-dimensional diagram of a multi-fibre ferrule with optical fibre stubs that is suitable for use within the connector device in FIG. 2.

(4) FIG. 4 presents the multi-fibre ferrule with optical fibre stubs in FIG. 3, said optical fibre stubs being spliced to the corresponding fibres of an optical cable.

(5) FIG. 5A is a perspective view of the splice protector for implementing the various aspects of the present invention.

(6) FIG. 5B presents a lateral view of the splice protector of FIG. 5A.

(7) FIG. 5C presents an end view of the splice protector of FIG. 5A.

(8) FIG. 5D presents the splice protector of FIG. 5A. It is shown aligned with the multi-fibre ferrule of FIG. 3.

(9) FIG. 5E presents the splice protector of FIG. 5A positioned on the splice site delimited between the optical fibre stubs of the multi-fibre ferrule of FIG. 3 and the corresponding optical fibres of an optical cable.

(10) FIGS. 6A through 6C display exemplary assembly steps for installing the splice protector of FIG. 5A on the multi-fibre splice site.

(11) FIG. 7A presents an assembly component for facilitating installation of the splice protector of FIG. 5A on a multi-fibre splice site.

(12) FIG. 7B presents the cover for the assembly component of FIG. 7A.

(13) FIGS. 8A and 8B display the assembly steps for internal components of the optical cable connector device of FIG. 2.

(14) FIG. 9A is a perspective view of another splice protector based on various aspects of the present invention.

(15) FIG. 9B is a perspective view of a part of the splice protector of FIG. 9A.

(16) FIG. 9C is a cross-sectional view of the splice protector of FIG. 9A.

(17) FIG. 9D presents the splice protector of FIG. 9A positioned on the splice site delimited between the optical fibre stubs of the multi-fibre ferrule of FIG. 3 and the corresponding optical fibres of an optical cable.

SPECIFIC EMBODIMENTS

(18) The guiding principles of the present disclosure relate to a splice-strengthening structure and

methods for its application. In certain embodiments, the splice-strengthening structure is constructed so as to effectively mount and protect a multi-fusion-splice site. Some examples of a splice-strengthening structure according to principles of the present invention have a low profile and are constructed so as to be easily mounted in a splice-on connector body. In certain embodiments, the splice-strengthening structure according to principles of the present invention has a low-profile splice protector sleeve. Said low-profile splice protector sleeve is used in combination with a curable splice protection material. Said curable splice protection material fills the sleeve and the splice packaged and positioned within the sleeve. In some examples, the splice protector sleeve has a U-shaped profile when viewed longitudinally. In some examples, the splice protector sleeve has long, thin, low-profile construction. It has a first side wall, a second side wall opposite the first side wall, a third wall connecting the first side wall and the second side wall, a longitudinal open side opposite the third wall, and open ends. In some embodiments, the splice protector sleeve comprises two or more matching parts (e.g., two halves). Every aspect of the present invention is applicable to hardened and non-hardened splice-on connectors.

(19) FIG. 1 and FIG. 2 show an exemplary, multi-aspect optical fibre connector device **1000** suitable for implementing the present invention. The optical fibre connector device **1000** comprises a hardened multi-fibre optical connector **20**. This multi-fibre optical connector **20** is optically coupled to a non-hardened multi-fibre optical connector **22** (e.g., an MPO connector) via an optical fibre adapter **24**. The optical fibre adapter **24** is constructed for installation within a housing or on a panel and is limited to receiving the hardened and sealed port of the hardened multi-fibre optical connector **20** and receiving the non-hardened port of the non-hardened multi-fibre optical connector **22**.

(20) Refer to FIG. 2. The hardened multi-fibre optical connector **20** is a splice-on connector coupled to an optical cable **400**. The optical cable **400** comprises multiple optical fibres **120** (e.g., single-core or multi-fibre ribbon) contained within a sheath **460**. The optical cable **400** further comprises strength members **461**. The strength members are for providing optical cable **400** stretch and compression reinforcement. As shown in the figure, the optical cable sheath **460** has a flat structure. In other embodiments, the optical cable may have other shapes (e.g., a round cable, a butterfly-shaped optical cable, etc.). In certain embodiments, the strength members **461** may comprise strengthening rods delimited by fiberglass reinforced epoxy resin. However, another type of strength member might be used (e.g., aramid fibre).

(21) The hardened multi-fibre optical connector **20** comprises a connector body **410**. The connector body **410** supports a multi-fibre ferrule **210** at the front end of the connector body **410**. The multi-fibre ferrule **210** is pressed upward, in a forward direction relative to the connector body **410**, by a spring. The hardened multi-fibre optical connector **20** further comprises a strengthening sleeve **26** and a housing **28** which fit over the connector body **410**. The housing **28** comprises a thin length **28a** that fits over the strengthening sleeve **26** and an end cap **28b** that fits over the multi-fibre ferrule **210**. A scaling part **30** may be disposed on the thin length **28a** and is for forming a seal with the optical fibre adapter **24** when the hardened multi-fibre optical connector **20** is secured in the hardened port of the optical fibre adapter **24**. The hardened fastener **32** is provided for securing the hardened multi-fibre optical connector **20** in the hardened port of the optical fibre adapter **24**. In the embodiment described, the fastener is a twist-lock fastener. It is shown as an external-thread nut. Said external thread binds with the corresponding internal thread delimited internally by the hardened port of the optical fibre adapter **24**. In another example, another type of twist-lock fastener may be used such as a snap fastener or alternatively an interior thread sleeve. The hardened multi-fibre optical connector **20** further comprises a shape-memory sleeve **34** (e.g., heat-shrink tubing), which provides a seal between the optical cable sheath **460** and the housing **28**. The hardened multi-fibre optical connector **20** further comprises a strain-relief sleeve **36**, which provides bending-radius protection and strain relief at the interface between the optical cable **400** and the housing **28**.

(22) The connector body **410** of the hardened multi-fibre optical connector **20** may comprise a main body **411**, a first cover part **412**, and a second cover part **413**. In the example shown, the first cover part **412** and the second cover part **413** are installed on the longitudinal-side opening **414** of the main body **411**. The cover part **412** may comprise a part that serves as a spring retainer. After the optical fibre stubs **110** (refer to FIGS. 3 and 4) supported by the ferrule **210** have undergone multiple fusion splicing at the splice site **130** with the optical fibres **120** of the optical cable **400**, the longitudinal-side opening **414** permits the ferrule **210** to be laterally installed into the main body **411**. In the example, the splice site **130** is protected by the splice protection system **100**. The splice protection system **100** comprises a splice protector sleeve **101** (refer to FIGS. 2, 6A, and 6B) and adhesive cement **109** (refer to FIG. 5B).

(23) FIG. 3 presents a multi-fibre ferrule assembly **200**. The multi-fibre assembly **200** comprises a ferrule **210**. The ferrule **210** supports multiple stubs having multiple optical fibres **110**. The ferrule comprises a front end **211**. The front end **211** is positioned opposite the back end **212**. The optical fibres **110** preferably are aligned within the ferrule in a row structure. The row structure of the optical fibres **110** has a width **W110**. The optical fibres **110** extend through the ferrule **210**, with the short lead-out end **111** extending out from the rear end **212** of the ferrule **210** and from the optical fibre **110** front ends **112** located at the front end **211** of the ferrule **210**. The front ends **112** of the optical fibres **110** are polished. It is possible to access the polished front ends **112** at the front end **211** of the ferrule **210**.

(24) FIG. 4 presents an optical cable **400**. The optical cable **400** has optical fibres **120** that are spliced to the optical fibres **110** of the ferrule assembly **200** at the splice site **130**. Generally, the ends of the exposed optical fibres **110** and **120** are fused together using an energy source (e.g., electric arc), and thus the optical fibres **110** and **120** are spliced together.

(25) The exposed part **140** of the optical cable **400** is preferably protected to prevent breaks. FIGS. 5A through 5E are diagrams of the splice protection system **100** according to the principles of the present disclosure. As shown in the figures, the splice protection system **100** comprises a splice protector sleeve **101**. The splice protector sleeve **101** has a width **W101**, a length **L101** delimited by the first end **107** and the second end **108**, and a thickness **T101** delimited by a first main side and a second main side. The first main side and the second main side comprise a first main-side wall **102** and a second main-side wall **103**. The first main-side wall **102** and the second main-side wall **103** are separated by an internal space **106** having the distance (i.e., thickness) **S106**. The main-side walls **102** and **103** may be roughly rectangular. The splice protector sleeve **101** further comprises first and second longitudinal secondary sides located opposite to each other and comprising a closed secondary wall **104** connected to the first side wall **102** and second side wall **103** and an open side **105**. The splice protector sleeve **101** further comprises open lateral secondary sides **107** and **108**. In the example, the width **W101** is greater than the thickness **T101**. The width **W101** may be, for example, at least three times as large as thickness **T101**, at least four times thickness **T101**, or at least five or six times its thickness. In the example shown, the splice connector sleeve **101** has a U-shaped cross-section (FIG. 5C). The U-shaped cross-section could be round, V-shaped, rectangular, or rectangular with rounded corners. The width **W101** of the splice protector sleeve **101** is preferably greater than the width **W110** of the optical fibres **110** row structure at the splice site **130**, so that when the splice site **130** is placed within the internal space **106** of the splice protector sleeve **101**, the first side wall **102** and second side wall **103** basically envelop the optical fibres **110** along the length of the splice protector sleeve **101**.

(26) FIGS. 6A through 6C show assembly and positioning of the splice protector sleeve **101** around the splice site **130**. Before splicing the optical fibres **120** of the optical cable **400** to the optical fibres **110** of the ferrule assembly **200**, a certain length of the sheath **460** is stripped away, and protective coating is removed from the optical fibres **120** to expose the optical fibres **120**. Then the exposed optical fibres **120** are passed through the central aperture in the spring **311** and an optional spring cover **312**, thus placing the spring **311** and the optional spring cap **312** over the optical cable

400. The optical fibres **110** and **120** preferably are spliced together by fusion splicing.

(27) According to one embodiment, before the exposed part **140** is placed into the splice protector sleeve **101**, adhesive cement **109** is injected into the internal space **106** of the splice protector sleeve **101**. In an alternative embodiment, the exposed part **140** is placed in the splice protector sleeve **101** prior to injection of the adhesive cement **109**. The adhesive cement is then injected into the internal space **106** of the splice protector sleeve **101**, filling the gaps between the optical fibres **110** and **120** and the splice protector sleeve **101**. In this manner, the adhesive cement packages the optical fibres and the splice site, thus stabilising and mechanically strengthening the splice site **130**.

(28) In an example, the adhesive cement is injected into the internal space **106** through an open side (e.g., the open longitudinal side **105**). In another embodiment, the splice protector sleeve **101** comprises at least one port **115** on a wall of the splice protector sleeve **101**. For example, one or more ports **115** could be positioned on the first main-side wall **102**, the second main-side wall **103**, and/or the secondary wall **104**. The adhesive cement **109** could be injected into the internal space **106** through the port **115**. In yet another embodiment, a cover **920** comprising a port **925** (refer to FIG. 7B) could be placed on the splice protector sleeve **101** so that it covers at least the open longitudinal side **105**. The adhesive cement **109** is injected through the port **925** into the splice protector sleeve **101**.

(29) During the process of assembling the splice protector sleeve **101** and the optical fibre cable **400** having the ferrule assembly **200**, these may optionally be retained in the mould **910** (refer to FIG. 7A). The mould **910** may comprise a suitably-sized slot **911** for receiving the splice protector sleeve **101**. The slot **911** preferably comprises one or more open sides allowing installation of the splice protector sleeve **101** into the mould **910**. The slot **911** may be shaped and sized so that, when the splice protector sleeve **101** is placed within the slot **911**, the open longitudinal side **105** of the splice protector sleeve **101** is plugged by part of the slot **911** and the side having the port **115** is exposed, which is for injecting the adhesive cement **109**. Alternatively, the splice protector sleeve **101** could be placed in the mould **910** with the open longitudinal side **105** facing upwards so that the adhesive cement **109** can be injected through the open side **105**. In an example, the mould **910** is suited to receiving a cover **920**. For example, the mould **910** could comprise a recess **912**. The cover **920** could sit within the recess **912**, with the result that, when the splice protector sleeve **101** is disposed within the mould **910**, the cover **920** seals the open longitudinal side **105** of the splice protector sleeve **101**, and the port **925** of the cover **920** lines up with the open longitudinal side **105** so that the adhesive cement **109** is injected into the splice protector sleeve **101** through the port **925**. The mould **910** further helps to hold the cover **920** firmly in a suitable position while the adhesive cement **109** is being injected.

(30) After the adhesive cement **109** is injected, the adhesive cement **109** could, for example be cured by UV radiation or by heating. The types of adhesive cement **109** may comprise UV-curable adhesive cement, heat-curable adhesive cement, or some other suitable adhesive cement. The types of adhesive cement **109** could comprise, for example, epoxy resin or some other type of resin (e.g., an acrylic resin such as cyanoacrylate, polyester resin, or some other suitable resin). In one embodiment, the splice protector sleeve **101** and the adhesive cement **109** provide a flexible splice protection system **100**. The flexible splice protection system **100** may be bent without causing damage. The splice protector sleeve **101** may be made from a polymeric material. For example, it may be made from polycarbonate (PC) or polyethyleneimine (PEI) or any other suitable material. In one embodiment, the splice protector sleeve **101** is made of material that can be penetrated by UV light. In one example, the adhesive cement **109** is shown as permanently securing the splice protector sleeve **101** onto the optical cable **400** and covering the exposed part **140**.

(31) After the splice protection system **100** has been installed on the splice site, the spring **311** and the spring cover **312** can slide forward over the splice protector sleeve **101** in the direction of the ferrule, bringing the spring cap **312** into contact with the rear end **212** of the ferrule **210**. In the example shown, the splice protector sleeve **101** is separate from the ferrule **210**. The ferrule **210**,

the spring **311**, the cover **312**, and the splice protector sleeve **101** containing the splice site **130** can be mounted in the main body **411** of the connector body **410** through the longitudinal-side opening **414** (see FIG. **8A**). The side opening **414** can then be covered by the first cover part **412** and the second cover part **413**, and the spring can be secured to the appropriate position by the spring retainer of the cover **412**. Adhesive cement can be used to fix the strength members **461** of the optical cable **400** to the rear end of the connector body **410**. Other external components of the hardened multi-fibre optical connector may be assembled onto the connector body **410**. Clearly, the strengthening sleeve **26**, the thin length **28a**, the fastener **32**, the shape-memory sleeve **34**, and the sleeve **36** can slide over the optical cable **400** before splicing and, moreover, slide back on the connector body **410** after the connector body **410** has been assembled.

(32) FIGS. **9A** through **9D** show an alternative embodiment of a splice protector sleeve **151** according to the principles of the present disclosure. As shown in FIG. **9A**, the splice protector sleeve **151** may consist of two half-shells **160** and **170**. These half-shells **160** and **170** are connected along longitudinal secondary sides. The splice protector sleeve **151** has a length **L151** delimited by a first end **152** and a second end **153**, a width **W151**, and a thickness **T151** delimited by a first main side **161** and a second main side **171** (refer to FIG. **9C**). In the example, the width **W151** is greater than the thickness **T151**. The width **W151** could be, for example, at least twice the thickness **T151** or at least three times the thickness **T151**, or four or five times. The splice protector sleeve **151** further comprises an internal space **156** having the distance **S156**. The splice protector sleeve **151** further comprises lateral secondary sides **157** and **158** located at the open first end **152** and second end **153**.

(33) The first half-shell **160** is described here and is shown in FIG. **9B**. The second half-shell **170** may have the same or similar (e.g., mirror) structure as the first half-shell **160**. The first half-shell **160** comprises a main-side wall **161** extending from the first end **162** to the second end **163** along the first longitudinal side **164** and the second longitudinal side **165**. The first half-shell **160** further comprises a first secondary wall extension **166** extending from the main-side wall **161** along the first longitudinal side **164** and a second secondary wall extension **167** extending from the main-side wall **161** along the second longitudinal side **165**. The first half-shell **160** may comprise one or more snap components, comprising one or more snap insertion holes **168** and one or more snap projecting blocks **169**. The snap components are suitable for connecting with the corresponding snap components on the second half-shell **170** and serve to secure the half-shells **160** and **170** together to form the splice protector sleeve **151**.

(34) When the half-shells **160** and **170** are in the assembled positions shown in FIG. **9D**, the snap components **168/179** and **169/178** connect together, and the closed seam **180** is delimited by the first and second secondary wall extensions **166/176** and **167/177**. When the splice site **130** is placed in the internal space **156** of the splice protector sleeve **151**, the first and second main-side walls **161/171** and the secondary wall extensions **166/176** and **167/177** basically envelope the optical fibres **110** along the splice connector sleeve **151**. In the embodiment shown, the splice connector sleeve **151** is separate from the ferrule **210**.

(35) In certain embodiments, when there is no permanent housing, the splice site between said multiple first optical fibres and multiple second optical fibres can be protected by coating materials (e.g., packaging materials such as adhesive cement). These materials could be UV-curable materials. A protective coating could be applied by spraying, injecting, or overmoulding, or by another technique. In certain examples, the protective coating materials could be injected or sprayed into, or otherwise fill, a mould around the splice site. Packaging materials could be cured within the mould, and then the mould could be removed from the packaging material. In certain embodiments, a splice site protected by sealing agent that lacks housing could be contained within the connector body of a vibration-resistant optical fibre connector of a type described above (e.g., the hardened multi-fibre connector **20**). This vibration-resistant optical fibre connector has a twist-lock fastener. This twist-lock fastener is for securing the vibration-resistant optical fibre connector

in the matching port of a vibration-resistant optical fibre adapter. A sealing part may be disposed between the vibration-resistant optical fibre connector and vibration-resistant optical fibre adapter. (36) Although some embodiments of the present invention have been described, other embodiments may exist. The particular features and actions described above are disclosed as illustrative aspects and embodiments of the present invention. After reading the descriptions herein, a person with ordinary skill in the art could become inspired with various other aspects, embodiments, modifications, and other equivalents without departing from the spirit of the present invention or the scope of the subject matter of the claims.

LIST OF PARTS

(37) **20** hardened multi-fibre connector **22** non-hardened multi-fibre optical connector **24** optical fibre adapter **26** strengthening sleeve **28** housing **28a** thin length **28b** cover **30** sealing part **32** fastener **34** shape-memory sleeve **36** strain-relief sleeve **100** splice protection system **101** splice protector sleeve **L101** length **W101** width **T101** thickness **102** first main-side wall **103** second main-side wall **104** closed secondary wall **105** open side **106** internal space **S106** distance **107** first end **108** second end **109** adhesive cement **110** optical fibres (stubs) **W110** width **115** port **120** optical fibres **130** splice site **140** exposed part **151** splice protector sleeve **T151** thickness **W151** width **155** port **156** internal space **S156** distance **157** first secondary side **158** second secondary side **160** first half-shell **161** first main wall **162** first end **163** second end **164** first longitudinal side **165** second longitudinal side **166** first secondary wall extension **167** second secondary wall extension **168** snap insertion hole **169** snap projecting block **170** second half-shell **171** first main wall **172** first end **173** second end **174** first longitudinal side **175** second longitudinal side **176** first secondary extension **177** second secondary extension **178** snap insertion hole **179** snap projecting block **180** seam **200** ferrule assembly **210** multi-fibre ferrule **211** front end **212** rear end **311** spring **312** spring cap **400** optical cable **410** connector body **411** main body **412** first cover part **413** second cover part **414** longitudinal slot **460** sheath **461** strength member **910** mould **911** slot **920** cover **925** port **1000** optical fibre connector device

Claims

1. A multi-fiber optical connector comprising: a plurality of stubs spliced to a row of cable fibers at a splice location; a flexible splice protector having only a first sidewall component and a separate second sidewall component, the first and second sidewall components extending in parallel along opposing sides of the row of the cable fibers in the direction of a length of the cable fibers and defining a length of the flexible splice protector, the first and second sidewall components extending across the splice location, the first and second sidewall components connected at a first longitudinal side and at a second longitudinal side along the length of the flexible splice protector, the flexible splice protector having a width between the first and second longitudinal sides and a thickness between the first and second sidewall components, the width being at least six times as large as the thickness; and a connector body housing the flexible splice protector and having a multi-fiber ferrule connected to the plurality of stubs on a first end and a cable of the row of cable fibers extending from a second end, wherein the first and second sidewall components are connected by snap components, wherein the flexible splice protector includes at least one port for injecting adhesive.
2. The multi-fiber optical connector of claim 1, wherein the multi-fiber optical connector further comprises a spring surrounding a portion of the length of the splice protector.
3. The multi-fiber optical connector of claim 1, wherein the flexible splice protector further comprises splice protection material between the first and second sidewall components surrounding gaps within the row of cable fibers at the splice location.
4. The multi-fiber optical connector of claim 1, wherein the flexible splice protector is made from polymeric material.

5. The multi-fiber optical connector of claim 1, wherein the first and second sidewall components are rectangular in shape.
 6. The multi-fiber optical connector of claim 1, wherein the length of the flexible splice protector is greater than the width.
 7. The multi-fiber optical connector of claim 1, wherein the flexible splice protector may be bent without causing damage.
 8. The multi-fiber optical connector of claim 1, wherein the flexible splice protector does not include a reinforcement member.
 9. The multi-fiber optical connector of claim 1, wherein the flexible splice protector does not require heating.
 10. The multi-fiber optical connector of claim 1, wherein the snap components are defined in secondary wall extensions of the first and second sidewall components, wherein the secondary wall extensions run adjacent to the first and second longitudinal sides.
 11. The multi-fiber optical connector of claim 1, wherein the first and second sidewall components are substantially structurally identical.
 12. The multi-fiber optical connector of claim 1, wherein the first and second sidewall components are first and second half-shells.
 13. A multi-fiber optical connector comprising: a plurality of stubs spliced to a row of cable fibers at a splice location; a flexible splice protector having a first sidewall and a separate second sidewall extending in parallel along opposing sides of the row of the cable fibers in the direction of a length of the cable fibers and defining a length of the flexible splice protector, the first and second sidewalls extending across the splice location, the first and second sidewalls connected at a first end and at a second end along the length of the flexible splice protector, the flexible splice protector having a width between the first and second ends and a thickness between the first and second sidewalls, the width being at least six times as large as the thickness; and a connector body housing the flexible splice protector and having a multi-fiber ferrule connected to the plurality of stubs on a first end and a cable of the row of cable fibers extending from a second end; wherein the multi-fiber optical connector further comprises a spring surrounding a portion of the length of the splice protector.
-